Kshitij Deshpande

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EDUCATION

Btech Integrated - MPSTME, NMIMS **2022 – Present** | Mumbai
CGPA: 3.59/4

TECHNICAL SKILLS

Languages:
Python, JavaScript, TypeScript, Kotlin, Shell Scripting

Libraries & Frameworks:
FastAPI, Flask, Node.js, Express.js, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn,

Cloud & DevOps:
Google Cloud Platform (GCP), Docker, Terraform, Grafana, Prometheus

Tools & Fundamentals:
Git, GitHub, Linux, SQLite, Supabase, Networking Fundamentals, Role-Based Access Control (RBAC)

AI/ML:
LLM Integration, AI Agent Workflows, TF-IDF, LinearSVC, Transformers.js, ONNX Runtime

Databases & Tools:
PostgreSQL, SQLite, Supabase, Git, GitHub, Linux, pytest, Playwright

Web & Security:
REST APIs, Browser Extensions (Manifest V3), PII Detection and Redaction, Role-Based Access Control (RBAC)

PROFESSIONAL EXPERIENCE

Shivaami Cloud Services Pvt. Ltd. **05/2026 – 07/2026** | Mumbai
Technical Intern

- **Terraform Autoscaling Infrastructure**: Provisioned autoscaling GCP infrastructure via Terraform, Global HTTP Load Balancer, Managed Instance Group, CPU-based autoscaler, and a custom Flask monitoring dashboard
- **Multi-VM Cloud Monitoring**: Built a Prometheus + Grafana observability stack across a custom GCP VPC (3 VMs, 2 subnets), tracking CPU, memory, disk, and network metrics
- **Windows Monitoring Stack**: Deployed a containerized Prometheus + Grafana monitoring stack on Docker with Windows Exporter for local system observability

PROJECTS

Automated Attendance System
Smart India Hackathon (Cleared Internal Round)
Built a facial recognition attendance platform (FastAPI, OpenCV) with liveness detection, Supabase Auth for role-based security, real-time sessions via Jitsi API, and a PowerBI analytics dashboard.

Agentic AI Support System
CyberCypher Hackathon
Architected a headless e-commerce migration agent (Flask, Gemini 2.5 Flash) with a custom ORDA (Observe-Reason-Decide-Act) loop and JSON policy engine to automate decision-making pipelines.

Reflect AI — Relationship Intelligence Dashboard
Engineered a real-time Telegram ingestion pipeline (Telethon, Flask) that chunks live conversation data and uses Llama-3.3-70B (Groq) for sentiment analysis and interaction health scoring, visualized on a dynamic dashboard.

Privacy-Preserving On-Device Vision Browser Agent
Built a Manifest V3 browser extension for webpage analysis and PII redaction before AI processing. Integrated DOM extraction, regex-based detection, Transformers.js, ONNX Runtime, WebGPU/WASM, and local Ollama models.

CyberShield AI — Cyberbullying & Profanity Detection Extension
Built a browser extension to detect abusive webpage content using a Flask API, word- and character-level TF-IDF, LinearSVC, and rule-based profanity detection. Added dynamic page scanning, configurable detection thresholds, local detection history, and automated tests.

Rasoi — AI-Powered Household Management Agent
Built an AI household agent using FastAPI, React, and PostgreSQL for personalized meal planning, pantry tracking, and budget-aware grocery management .Implemented checkpointed agent workflows, dietary rule validation, human approval, voice input, payment and delivery integrations, and idempotent crash recovery.